Conceptual Clarity is the Basis for Advancing a Research Field

Peer Ecologies and Classroom Heterogeneity

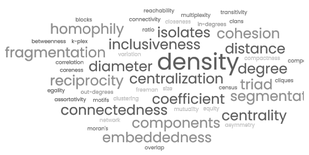
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PROJECT 1: PEER ECOLOGY

Conceptualizing Classroom Peer Ecologies

• Classroom Microsystem: Defined as 'the ultimate mechanism through which development occurs' (Bronfenbrenner, 1994, 1996; Gest & Rodkin, 2011).







DIFFERENT ASPECTS OF PEER ECOLOGY

Hierarchy
- Centralization

Categorization norms
- Assortativity

Reciprocity
- Reciprocity

Local clustering
- Shared mutual acquaintances

Core-periphery structure
- Centralization
- Clique size
- Embeddedness

Social integration
- Density
- Reciprocity
- Transitivity

Status orientation
- Indegree popularity

Connectivity and inclusiveness
- Component ratio
- Structural cohesion
- Number of isolates

Popularity norms
- r(indegree popularity, attribute)

Subgroup distincitveness
- Relative density ratio
- Modularity
- Structural cohesion

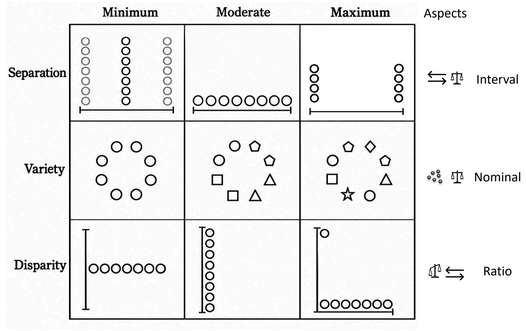
Density and cohesion
- Density
- Segmentation
- Diameter
- Compactness

Homophily
- Assortativity
- Moran's I

Gest et al., 2011
Doehne et al., 2025
Neuendorf et al., 2026

PROJECT 2: CLASSROOM HETEROGENEITY

- „In the most general sense, classroom heterogeneity is understood as the presence of individual differences within a group of students.“
(Neuendorf et al., 2026)
- Necessary specifications:
  - Level, (e.g., classroom, grade level, school)
  - Dimension (e.g., SES, ethnicity, gender, achievement)
  - Type (e.g., separation, variety, disparity, richness)



Harrison & Klein, 2007

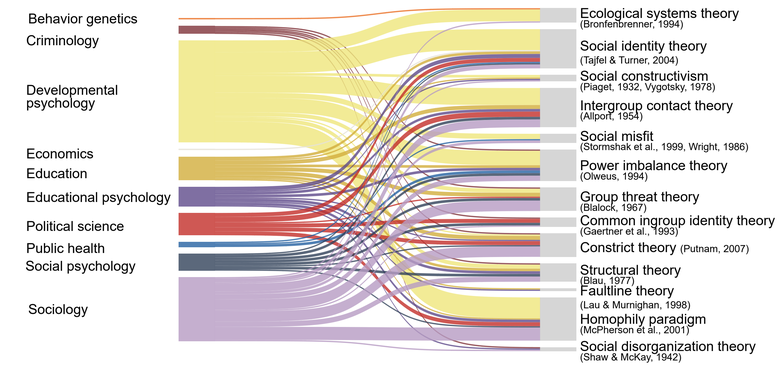
CONCEPTUAL CLARITY WITHIN THEORY DEVELOPMENT

- Putting more emphasis on sound concepts is needed to advance both theory development and knowledge in a field.
- In peer ecology research, a multitude of indicators are used, but no typology of classrooms exists.
- In classroom heterogeneity research, theories imply different heterogeneity concepts, but researchers use a single one without reflecting its alignment with the cited theories.
- Each empirical study should go back to the basics of describing the concepts in relation to the empirical phenomena under study, to discuss conceptual ambiguities and to justify the measurement approaches (Bringmann et al., 2022).

DISCONNECT BETWEEN THEORY AND MEASUREMENT

Review: Classroom Heterogeneity and Psycho-Social Student Outcomes

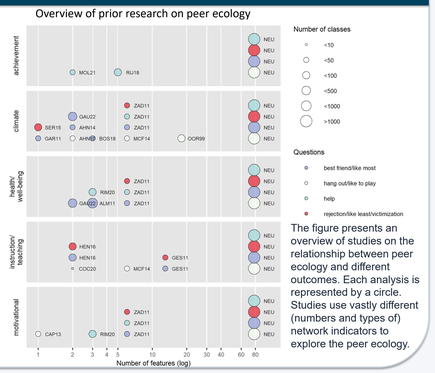
- Reseach disciplies cite diverse theories.



- Theoretical mechanisms are based on different understanding of heterogeneity.

HOW TO ADVANCE THIS FIELD?

- Issue: Peer ecology as a whole lacks conceptual clarity.
- Network indicators are interrelated in complex ways. Unit-centred approaches are promising.
- Solution: Combining theory and data-driven dimensions of peer ecology for typology and theory development.



The figure presents an overview of studies on the relationship between peer ecology and different outcomes. Each analysis is represented by a circle. Studies use vastly different (numbers and types of) network indicators to explore the peer ecology.

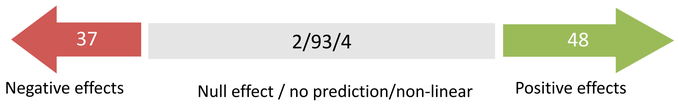
FUNDING PROPOSAL

Peer Ecologies and Classroom Composition

1) Conceptual part: What is Peer Ecology?
   - develop an integrated conceptualization of peer ecology which takes into account theoretical discussions and empirical approaches from prior research on the topic
   - systematic review
2) Empirical part I: How is it related to classroom composition?
   - characterize types of classrooms exhibiting distinct profiles across dimensions of peer ecology,
   - analyze relationships between aspects of classroom composition and peer ecology profiles
   - cross-sectional analysis using multiple large-scale data sets
3) Empirical part II: Do changes in composition lead to changes in peer ecology?
   - longitudinal analysis using multiple large-scale data sets

SIGNS OF THEORETICAL IMMATURITY

- Most research finds it difficult to state **directed hypotheses**.



- Mostly inconclusive results.
- Weak discussion about theory – concept – measurement link.
- Prevailing approach: measurement → concept | theory.

Bringmann, L. F., Elmer, T., & Eronen, M. I. (2022). Back to Basics: The Importance of Conceptual Clarification in Psychological Science. Current Directions in Psychological Science, 31(4), 340–346. https://doi.org/10.1177/09637214221096485
Bronfenbrenner, U. (1994). Ecological Models of Human Development. In P. Peterson, E. Baker, & B. McGaw (Eds.), International Encyclopedia of Education (2nd ed., pp. 1643–1647). Elsevier.
Doehne, M., McFarland, D. A., & Moody, J. (2024). Network ecology: Tie fitness in social context(s). Social Networks, 76, 174–190. https://doi.org/10.1016/j.socnet.2023.09.005
Gest, S. D., & Rodkin, P. C. (2011). Teaching practices and elementary classroom peer ecologies. Journal of Applied Developmental Psychology, 32(5), 288–296. https://doi.org/10.1016/j.appdev.2011.02.004
Harrison, D. A., & Klein, K. J. (2007). What's the Difference? Diversity Contructs as Separation, Variety, or Disparity in Organizations. The Academy of Management Review, 32(4), 1199–1228.
Neuendorf, C., Hascher, S., Hou, C., & Rjosk, C. (n.d.). School and Classroom Heterogeneity and Psychosocial Student Outcomes: Mapping an Emerging Interdisciplinary Research Field. Retrieved from osf.io/preprints/psyarxiv/3kuby_v1
Neuendorf, C., Masuda, N., & Murayama, K. (2026). The global structure of classroom social networks predicts academic and psycho-social adjustment. PNAS Nexus, Article pgag291. Advance online publication. https://doi.org/10.1093/pnasnexus/pgag291